EE2005: Electrical Machines
Tutorial 3

1. A 200 V/400 V, 1-phase, 50 Hz, 4 kVA transformer has the following equivalent circuit parameters referred to the low-voltage side: $R = 0.15\,\Omega$, $X = 0.37\,\Omega$, $R_m = 600\,\Omega$, $X_m = 300\,\Omega$. For a high voltage current of 10 A at a lagging power factor of 0.8, calculate (a) the low-voltage input current (b) the efficiency and (c) the voltage at the terminals of the high voltage side.

2. In a 25 kVA, 2000 V/200 V, 1-phase transformer, the iron and full-load copper losses are equal to 350 and 400 W respectively.

   (a) Calculate the efficiency when the unit supplies at UPF

   (i) full load

   (ii) half the full load

   (b) Determine the load for maximum efficiency and the iron and copper losses in this case.

3. The resistances and leakage reactances of a 10 kVA, 50 Hz, 2200 V/220 V distribution transformer are as follows: $R_1 = 4\,\Omega$, $R_2 = 0.04\,\Omega$, $X_1 = 5\,\Omega$ and $X_2 = 0.005\,\Omega$

   (a) Find the total leakage impedance referred to (i) HV side (ii) LV side.

   (b) If the transformer is supplied from an AC source of 2200 V at the primary side, and a full load current at 0.8 pf lag flows in the secondary, find the terminal voltage across the secondary.

   (c) Find the same as in part (b) for a full-load pf of 0.8 leading.

   (d) Considering the core losses to be 80 W, find the efficiency under condition given in part (b) and part (c).

   (e) If the load in part (b) gets short-circuited, find the steady-state current in the HV lines, assuming that the voltage applied to the transformer remains unchanged.

4. The nameplate on a 50 MVA, 60 Hz single-phase transformer indicates that it has a voltage rating of 8 kV/78 kV. An open-circuit test is conducted from the low-voltage side, and the corresponding instrument readings are 8 kV, 62.1 A, and 206 kW. Similarly, a short-circuit test from the low-voltage side gives readings of 674 V, 6.25 kA, and 187 kW.

   (a) Calculate the equivalent circuit parameters of the trans- former as referred to the low-voltage terminals.

   (b) Calculate the equivalent circuit parameters as referred to the high- voltage terminals.

   (c) Determine the efficiency and voltage regulation if the transformer is operating at the rated voltage and load (unity power factor).

   (d) Determine the load at UPF for which the efficiency is maximum.

5. The following data were obtained for a 20 kVA, 50 Hz, 2400 V/240 V distribution transformer when tested:
   With HV winding open circuited: 240 V, 1.006 A, 126.6 W
   With LV side short circuited: 57.5 V, 8.34 A, 284 W

   (a) Compute the efficiency at full load current and the rated terminal voltage at 0.8 power factor lagging.

   (b) Assume that the load power factor is varied while the load current and secondary terminal voltages are held constant. Use phasor diagram to determine the load power factor for which the regulation is highest. What is this regulation?